Graph Traversal.

~~Traversal~~ It is a technique used for searching a vertex in a graph. Also used to decide the order of ~~vertices~~ vertices. is visited in the search process. Finds the edges to be used in such process. ~~Without~~ without creating loops.

→ 2 Types of traversal : i) DFS (Depth first search)
ii) BFS (Breadth first search).

⇒ i) BFS :

→ BFS Traversal of ~~a~~ a graph produces a spanning tree. as a final result. spanning tree is a graph without loops or cycles.

→ use queue data structure with maximum size of total number of vertices in the graph to implement the BFS traversal

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Steps :-

Step 1 :- define a queue of size = total number of vertices in the graph.

Step 2 :- Select any vertex ~~has~~ as starting point for traversal. Visit that vertex and insert it into the queue.

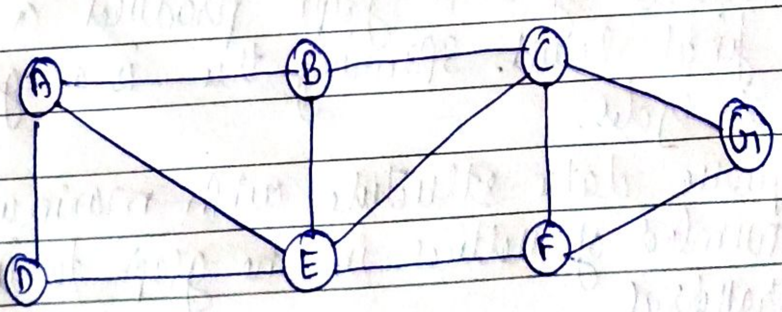
Step 3 :- Visit all the non visited adjacent vertices of the vertex which is at front of the queue and insert them into the queue.

Step 4 :- When there is no new vertex to be visited from the vertex which at the front of the queue then delete that vertex.

Step 5 :- Repeat step 3 and 4 until queue becomes empty.

Step 6 :- When queue becomes empty then produce final spanning tree by removing unused edges from the graph.

⇒



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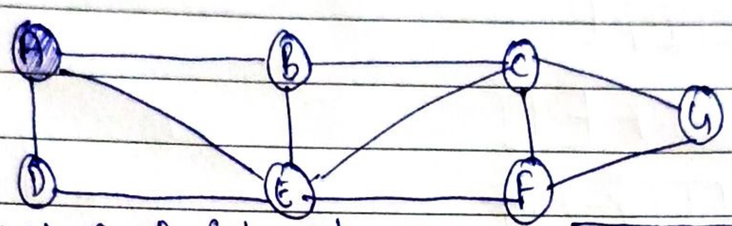
- Step 1 :-

queue

0	1	2	3	4	5	6

Select the vertex A as starting point (visit A)

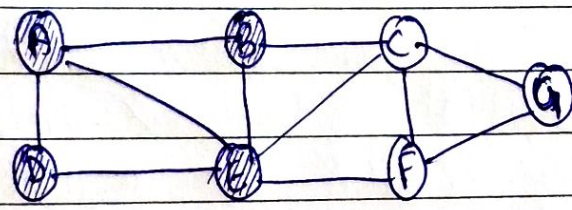


Insert A into the queue

A						

- Step 2 :-

Visit all adjacent vertices of A which are not visited before. (B, E, D)



note :- order is not imp while inserting to queue

Insert newly visited nodes D E B.

Queue

A	D	E	B			

• there is no new node to visit from A so delete A.

	D	E	B			

- Step 3 :-

Visit all adjacent vertices of D (Front of queue) which are not visited (there is no vertex).
delete D from the queue.

Queue

		E	B			